ARCHITECTURE & ENGINEERING SPECIFICATION • PHASE 3 AUTONOMOUS AI

Enterprise Inventory & Order Management System

Complete REST API Specification, AI Predictive & Autonomous Architecture & Page Flow Directory

SYSTEM ARCHITECTURE

Clean Architecture / DDD / Autonomous AI Suite

BACKEND FRAMEWORK

.NET 10.0 Web API / Entity Framework Core

FRONTEND APPLICATION

Angular Standalone / Signals & Reactive Forms

SECURITY MODEL

JWT Bearer Authentication & RBAC (4 Roles)

DATABASE & ACID GUARANTEES

SQL Server with Distributed Concurrency

RELEASE VERSION

v3.0.0 (Phase 3 Autonomous AI & Operations Release)

1. System Overview & Architecture

The **Enterprise Inventory and Order Management System** is an enterprise-grade platform designed for centralized stock tracking across multiple physical warehouses, supplier lifecycle administration, order fulfillment with ACID transactional stock reservations, and automated audit logging.

High-Level System Topology



Role-Based Access Control (RBAC) Matrix

Role	Access Scope	Permitted Operations
Admin	Full System Administration	Manage all users, view audit logs, create/update/delete products, configure warehouses, initiate transfers & adjustments, override order states.
WarehouseManager	Warehouse & Stock Operations	Perform physical stock adjustments, transfer stock between warehouses, manage product catalog (Create/Update), update order status to Processing/Completed.
SalesRepresentative	Sales & Customer Orders	Browse live product inventory across warehouses, create customer sales orders with real-time stock deduction, track order fulfillment, cancel orders.
Auditor	Compliance & Quality Control	Read-only inspection of inventory valuation, warehouse levels, stock movement transactions, and full immutable system audit logs.

2. Complete REST API Specification

Base URL: `http://localhost:5000/api` • Authentication: `Authorization: Bearer <token>`

2.1 Authentication & User Endpoints

POST `/api/auth/login` `Anonymous`

Validates credentials using BCrypt hash comparison, records user login audit log, and issues a 12-hour signed JWT token with identity and role claims.

Request Body:

```
{
  "username": "admin",
  "password": "Password123!"
}
```

Response (200 OK):

```
{
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "expiresAt": "2026-10-23T23:59:59Z",
  "user": {
    "id": 1,
    "username": "admin",
    "email": "admin@enterprise.com",
    "fullName": "Enterprise Administrator",
    "role": "Admin",
    "isActive": true
  }
}
```

GET `/api/auth/me` `Authenticated`

Returns the authenticated user profile based on the JWT bearer token claims.

Response (200 OK): Returns `UserDto` object.

GET `/api/auth/users` `Admin` `Auditor`

Retrieves list of all registered system users and assigned roles.

2.2 Dashboard KPI Endpoints

`GET` **`/api/dashboard/summary`** `Authenticated`

Computes executive KPIs including total active products, warehouse count, total inventory asset valuation (cost basis), total sales revenue, low stock count, and recent stock activity.

Response (200 OK):

```
{
  "totalProducts": 142,
  "totalWarehouses": 4,
  "lowStockItemCount": 3,
  "totalStockInventoryValue": 542980.50,
  "totalSalesRevenue": 128450.00,
  "recentMovements": [...]
}
```

2.3 Products Management Endpoints

Method	Route	Authorized Roles	Description
GET	`/api/products`	All Authenticated	Search & paginate catalog. Query params: `searchTerm`, `categoryId`, `supplierId`, `pageNumber`, `pageSize`.
GET	`/api/products/{id}`	All Authenticated	Fetch single product with category and supplier metadata.
POST	`/api/products`	Admin, WarehouseMgr	Create product with SKU validation, safety reorder thresholds, and unit price.
PUT	`/api/products/{id}`	Admin, WarehouseMgr	Update product details, cost price, and safety stock levels.
DELETE	`/api/products/{id}`	Admin	Soft delete product (enforces referential integrity check against orders).

2.4 Multi-Warehouse Inventory & Stock Movement Endpoints

Method	Route	Authorized Roles	Description
GET	`/api/inventory`	All Authenticated	Paginated inventory matrix across warehouses. Supports `lowStockOnly=true` filter.
GET	`/api/inventory/low-stock-alerts`	All Authenticated	Returns all inventory records where `QuantityOnHand <= ReorderPoint`.
POST	`/api/inventory/adjust`	Admin, WarehouseMgr	Adjust physical stock with reason audit (Cycle count, Damaged, Replenishment).
POST	`/api/inventory/transfer`	Admin, WarehouseMgr	ACID inter-warehouse transfer: atomically deducts source warehouse and credits target warehouse.
GET	`/api/inventory/movements`	All Authenticated	Historical audit ledger of all stock movement events (IN, OUT, TRANSFER, ADJUST).

> **ACID Concurrency Guarantee:** Inventory transfers execute within an explicit database transaction isolation scope. If the source warehouse lacks sufficient available stock or any database failure occurs, the entire transaction is rolled back with zero partial deduction.

2.5 Orders & Fulfillment Endpoints

`POST` **`/api/orders`** `Admin` `SalesRepresentative` `WarehouseMgr`

Atomically places a customer sales order and deducts available warehouse inventory in a single ACID transaction. If stock is insufficient, returns `400 Bad Request` with specific product availability.

Request Body:

```
{
  "customerName": "Acme Logistics Inc.",
  "customerEmail": "orders@acme.com",
  "shippingAddress": "100 Industrial Parkway, Chicago IL",
  "warehouseId": 1,
  "items": [
    { "productId": 4, "quantity": 10, "unitPrice": 49.99 }
  ]
}
```

Method	Route	Authorized Roles	Description
GET	`/api/orders`	All Authenticated	Paginated orders listing with customer name, status, and order total.
GET	`/api/orders/{id}`	All Authenticated	Fetch order details including individual line items and fulfillment history.
PUT	`/api/orders/{id}/status`	Admin WarehouseMgr Sales	Advance order status through lifecycle: `Pending → Processing → Completed`.
POST	`/api/orders/{id}/cancel`	Admin Sales WarehouseMgr	Cancels order and **automatically replenishes** deducted inventory back to warehouse stock.

2.6 Suppliers, Categories, Warehouses & Audit Endpoints

Method	Route	Authorized Roles	Description
GET	/api/categories	All Authenticated	List all product categories (Electronics, Apparel, Equipment, etc.).
POST	/api/categories	Admin, WarehouseMgr	Create new product category.
GET	/api/suppliers	All Authenticated	List suppliers with contact info, lead times, and reliability metrics.
POST	/api/suppliers	Admin, WarehouseMgr	Add new vendor profile with procurement terms.
GET	/api/warehouses	All Authenticated	List active distribution centers with capacity thresholds.
GET	/api/audit	Admin, Auditor	Query immutable audit trail records with before/after state diffs and IP tracking.

2.7 Phase 1: AI Document Ingestion & Copilot Endpoints

Method	Route	Authorized Roles	Description
POST	/api/ai/ingest-document	Admin, WarehouseMgr, Sales	Multimodal OCR document extraction for POs and invoices with catalog SKU reconciliation.
GET	/api/ai/document-samples	All Authenticated	Retrieve pre-seeded commercial invoices and PO demo documents for 1-click intake testing.
POST	/api/ai/copilot/query	All Authenticated	Natural language conversational AI queries over orders, stock, runways, and suppliers.

2.8 Phase 2: Predictive Demand Forecasting & Restock Endpoints

Method	Route	Authorized Roles	Description
GET	`/api/inventory/forecasting/summary`	All Authenticated	AI KPI summary: critical stockouts, restock recommended items, and overstock tied-up capital.
GET	`/api/inventory/forecasting/recommendations`	All Authenticated	Dynamic Safety Stock, Dynamic ROP, 30-day burn rate, and suggested replenish order quantities.
GET	`/api/inventory/forecasting/overstock`	All Authenticated	Detect stagnant >60-day carrying inventory with exact tied-up capital & rebalance recommendations.
GET	`/api/inventory/forecasting/product/{{id}}`	All Authenticated	Single-item runway prediction, burn rate, and projected stockout date.
POST	`/api/inventory/forecasting/generate-po-draft`	Admin WarehouseMgr	Batch restock PO generator aggregating recommended quantities and supplier lead times.

2.9 Phase 2: AI Catalog Intelligence & Deduplication Endpoints

Method	Route	Authorized Roles	Description
POST	`/api/products/intelligence/detect-duplicates`	Admin WarehouseMgr	Tokenized Jaccard similarity and normalized SKU check on product title input.
POST	`/api/products/intelligence/suggest-category`	Admin WarehouseMgr	AI automatic category prediction based on product title and technical specifications.
GET	`/api/products/intelligence/audit`	Admin WarehouseMgr	Full catalog hygiene scan returning duplicate pairs, category misclassifications, and quality score.

2.10 Phase 3: Intelligent Audit & Shrinkage / Anomaly Radar Endpoints

Method	Route	Authorized Roles	Description
GET	`/api/audit/anomalies/radar`	All Authenticated	Unsupervised anomaly radar scanning off-hours adjustments, abnormal cancellations, rapid shrinkage spikes, and transfer discrepancies with financial exposure scoring.
POST	`/api/audit/anomalies/resolve`	Admin Auditor WarehouseMgr	Acknowledge and resolve flagged security/shrinkage anomaly with supervisor audit notes and timestamped sign-off.

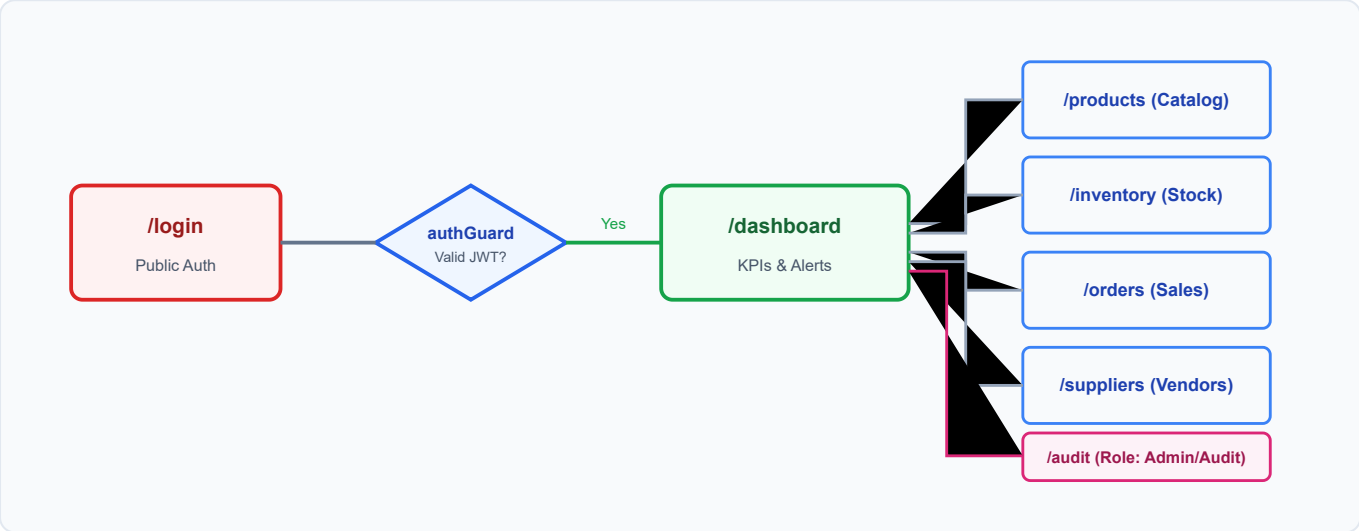
2.11 Phase 3: Supplier Reliability & Autonomous Procurement Policy Endpoints

Method	Route	Authorized Roles	Description
GET	`/api/suppliers/intelligence/scorecards`	All Authenticated	Evaluates supplier performance: Reliability Index, Tier categorization (🥇 Gold, 🥈 Silver, ⚠️ High Risk), lead-time variance, on-time probabilities, and failover alternate vendors.
GET	`/api/suppliers/intelligence/autonomous-policies`	All Authenticated	List all active autonomous replenishment policies governing automated PO drafting triggers and budget caps.
POST	`/api/suppliers/intelligence/toggle-policy`	Admin WarehouseMgr	Toggle autonomous replenishment policy on/off (pause or resume self-executing purchasing engine).

3. Frontend Application Architecture & Page Flow

The client single-page application is structured with Angular Standalone components, responsive navigation shell, client-side route guards, and reactive state management.

3.1 User Journey & Navigation Flow



3.2 Page-by-Page Specifications & Interaction Workflows

1. Login Page (`/login`)

- **Purpose:** Authenticate user and initialize secure session.
- **Features:** Form validation, quick-fill role buttons (Admin, Warehouse Mgr, Sales Rep, Auditor), authentication progress indicator.
- **Post-Login Navigation:** On HTTP 200, saves token to `localStorage`, populates user signal in `AuthService`, and redirects immediately to `/dashboard`.

2. Operations Dashboard (`/dashboard`)

- **Purpose:** Executive command center for inventory status, predictive alerts, and operational health.
- **KPI Cards:** Total Active Products, Inventory Asset Valuation ($), Low-Stock Items Count, Active Warehouses Count, Cumulative Sales Revenue ($).
- **Phase 2 AI Predictive Card:** Surfaces critical stockout risks (<3 days), restock recommended counts, overstocked items, and exact tied-up carrying capital ($).
- **Quick Actions:** Primary triggers for `+ New Order`, `Adjust Stock`, and `Explore Restock Recommendations`.

3. Products Catalog (`/products`)

- **Purpose:** Central product registry and AI catalog hygiene control.
- **Interactions:** Real-time text search (SKU/Title), Category dropdown filter, Supplier dropdown filter.
- **Phase 2 AI Catalog Deduplication:** Global 1-click catalog scanner identifying duplicate SKUs, Jaccard token overlap, and category misclassifications with overall Quality Score.
- **Phase 2 AI Auto-Pick:** Natural language semantic classifier suggests optimum product category automatically when creating/editing products.
- **Modals:** "Add Product", "Edit Product", and "AI Catalog Deduplication Audit" dialogs with live validation.

4. Multi-Warehouse Inventory (`/inventory`)

- **Purpose:** Physical stock visibility, AI predictive replenishment, and inter-warehouse movement orchestration.
- **Phase 2 AI Demand Forecasting Tab:** Analyzes 30-day order velocity to compute Dynamic Safety Stock (`SS = ⌈1.65 × √(Lead Time) × max(1, Daily Burn × 0.7)⌉`) and Dynamic Reorder Point (`ROP = (Daily Burn × Lead Time) + SS`).
- **Phase 2 Restock PO Generator:** 1-click batch Purchase Order builder with item breakdown, estimated costs, lead times, and instant replenishment execution.
- **Phase 2 Overstock & Dead Stock View:** Identifies capital tied up in excess items (>60 days runway) with 1-click rebalancing transfer actions.
- **Warehouse Selector:** Filter stock and forecasts by physical distribution facility.
- **Stock Adjustment & Transfer Modals:** Allows warehouse staff to modify quantities or transfer inventory with ACID transactional atomicity.
- **Movement Ledger:** Historical log of every stock alteration with timestamp, delta, and operating user.

5. Order Management & Fulfillment (`/orders`)

- **Purpose:** Sales order intake, AI invoice/PO document processing, and delivery tracking.
- **Phase 1 AI Document Ingestion:** Multimodal OCR scanner supporting PDF, PNG, and invoice images with automated line-item token parsing and catalog SKU matching.
- **Status Workflow:** Visual pill badges indicate state: `PENDING` → `PROCESSING` → `COMPLETED` or `CANCELLED` .
- **Order Creation Dialog:** Dynamic line-item builder with customer details. Automatically verifies warehouse stock levels before submission.
- **Order Cancellation Flow:** Prompts for cancellation reason and triggers the automatic inventory restock routine on the backend.

6. Suppliers Directory & Autonomous Procurement (`/suppliers`)

- **Purpose:** Vendor relationship management, reliability evaluation, and autonomous replenishment policies.
- **Phase 3 AI Supplier Scorecards:** Evaluates historical lead-time variance (±days), on-time delivery probability, and assigns vendor tiers (🥇 Gold Preferred, 🥈 Silver, ⚠️ High Risk).
- **Phase 3 Failover Routing:** Recommends alternate secondary vendors to mitigate supply-chain disruptions.
- **Phase 3 Autonomous Procurement Policies:** Interactive self-governing replenishment rules that automatically draft purchase orders when runway breaches safety boundaries (≤ threshold days, PO cap limits).
- **Directory Actions:** Add new vendor profiles, edit terms, and filter supplied products.

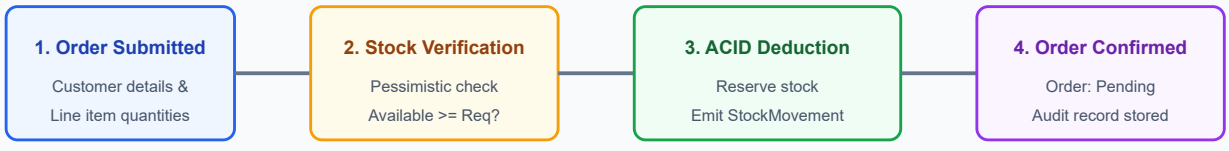
7. System Audit Trail & AI Shrinkage Radar (`/audit`)

- **Purpose:** Compliance oversight, regulatory traceability, and automated shrinkage detection (restricted to Admin, Auditor, and Warehouse Managers).
- **Phase 3 AI Shrinkage & Anomaly Radar:** Unsupervised heuristic anomaly scanner evaluating off-hours stock adjustments, abnormal cancellations, rapid shrinkage spikes, and transfer discrepancies.
- **Executive Risk Metrics:** Overall Network Risk level, Active Flagged Anomalies, Critical Exposure ($), and Discrepancy Units.
- **1-Click Investigation & Resolution:** In-app supervisor audit modal allowing warehouse supervisors to verify physical counts, record resolution notes, and resolve security flags.
- **Compliance Audit Logs Tab:** Immutable record of user actions, payload diffs, and IP addresses.

8. Global Enterprise AI Copilot (Phase 1, 2 & 3 Autonomous AI)

- **Purpose:** Conversational intelligence assistant accessible across all screens via slide-out drawer.
- **Capabilities:** Answers plain-English inquiries about shrinkage anomalies, supplier reliability scorecards, autonomous replenishment, predictive demand forecasting, dynamic restock runway, overstock capital, warehouse valuations, and catalog deduplication.
- **Deep Navigation:** Generates interactive action buttons directly navigating to relevant application views and opening workflows (e.g., `/audit?tab=anomalies`, `/suppliers?view=ai-scorecard`, `/inventory?tab=forecast`).

3.3 Order Fulfillment & Stock Deduction Lifecycle



4. Error Handling & Architectural Standards

Standard RFC 9110 Error Response Structure

All HTTP exception scenarios (validation faults, missing resources, unauthorized attempts, concurrency conflicts) return a structured JSON response managed by the centralized `ExceptionHandlingMiddleware`:

```
{
  "status": 400,
  "title": "Insufficient Stock",
  "detail": "Requested 50 units of Wireless Keyboard, but only 12 are available in Central Warehouse.",
  "errors": {
    "productId": 4,
    "productName": "Wireless Keyboard",
    "availableQuantity": 12,
    "requestedQuantity": 50
  },
  "timestamp": "2026-10-23T06:45:00.123Z"
}
```

HTTP Status Code Mapping

Status Code	Title	Trigger Scenario
200 OK	Success	Query succeeded or entity updated.
201 Created	Resource Created	Product, category, order, or supplier created with `Location` header.
204 No Content	No Content	Deletion completed successfully or preflight response.
400 Bad Request	Validation / Business Rule	Insufficient inventory, invalid SKU, duplicate identifier.
401 Unauthorized	Authentication Missing	Missing, expired, or invalid JWT Bearer token.
403 Forbidden	Insufficient Role	Authenticated user does not possess required role claim.
404 Not Found	Resource Not Found	Target entity ID does not exist in the database.
409 Conflict	Concurrency Conflict	Simultaneous modification detected on same inventory row.

> **Security Note:** Passwords are never stored or logged in plain text. All user passwords are salted and hashed using BCrypt. JWT secret keys are stored securely in backend configuration with minimum 256-bit cryptographic strength.

— End of Technical Specification Document —